APPROXIMATING A RELATIVE MAP THAT IS AN ISOMORPHISM ON COMPLETIONS BY AN ETALE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 11720-11725 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 11727-11798, the complete author proof (70 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The relative map approximation lemma, completion of local rings, elementary etale neighbourhoods, and the lifting criterion for smoothness; the reduction is retained.

Source reference. Original lemma algebra-lemma-surjective-endo-noetherian-ring-is-iso: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11798–11798. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-control-graded: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11571–11590. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-lifting-along-artinian-at-point: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3575–3600. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-relative-map-approximation: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11289–11363. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma more-algebra-lemma-flat-unramified: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11798–11798. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-etale-smooth-unramified: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11798–11798. Cited by

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *With notation an assumptions as in Lemma lemma-relative-map-approximation assume that φ induces an isomorphism on completions. Then we can choose our diagram such that f is étale.*

Proof. We may assume $N \geq 2$ and we may replace (T, t) with (T', t') as in Lemma lemma-control-graded. Since $(V, v) \to (S, s)$ is an elementary étale neighbourhood, so is $(X \times_S V, (x, v)) \to (X, x)$. Thus $\mathcal{O}_{X,x} \to \mathcal{O}_{X \times_S V,(x,v)}$ induces an isomorphism on completions by More on Algebra, Lemma more-algebra-lemma-flat-unramified. We claim $\mathcal{O}_{X,x} \to \mathcal{O}_{W,w}$ induces an isomorphism on completions. Having proved this, Lemma lemma-lifting-along-artinian-at-point will show that f is smooth at w and of course f is unramified at u as well, so Morphisms, Lemma morphisms-lemma-etale-smooth-unramified tells us f is étale at w.

First we use the commutativity in part (5) of Lemma lemma-relative-map-approximation to see that for $i \leq N$ there is a commutative diagram

$$\begin{array}{ccccc}
\mathrm{Gr}^i_{\mathfrak{m}_x}(\mathcal{O}_{X,x}) & \xrightarrow{\varphi} & \mathrm{Gr}^i_{\mathfrak{m}_{y_\sigma}}(\mathcal{O}^\wedge_{Y_\sigma,y_\sigma}) & = & \mathrm{Gr}^i_{\mathfrak{m}_{(y,v)}}(\mathcal{O}_{Y \times_T \tau V,(y,v)}) \\
\| & & & & \cong \downarrow \\
\mathrm{Gr}^i_{\mathfrak{m}_x}(\mathcal{O}_{X,x}) & \xrightarrow{\cong} & \mathrm{Gr}^i_{\mathfrak{m}_{(x,v)}}(\mathcal{O}_{X \times_S V,(x,v)}) & \xrightarrow{f^\sharp_w} & \mathrm{Gr}^i_{\mathfrak{m}_w}(\mathcal{O}_{W,w})
\end{array}$$

This implies that $f^\sharp_w$ defines an isomorphism $\kappa(x) \to \kappa(w)$ on residue fields and an isomorphism $\mathfrak{m}_x/\mathfrak{m}_x^2 \to \mathfrak{m}_w/\mathfrak{m}_w^2$ on cotangent spaces. Hence $f^\sharp_w$ defines a surjection $\mathcal{O}^\wedge_{X,x} \to \mathcal{O}^\wedge_{W,w}$ on complete local rings.

By Lemma lemma-control-graded there is an isomorphism of $\mathrm{Gr}_{\mathfrak{m}_s}(\mathcal{O}_{(Y \times_T \tau V,(y,v)})$ with $\mathrm{Gr}_{\mathfrak{m}_s}(\mathcal{O}_{Y_\sigma,y_\sigma})$. This follows by taking stalks of the isomorphism of conormal sheaves at the point y. Since our local rings are Noetherian taking associated graded with respect to $\mathfrak{m}_s$ commutes with completion because completion with respect to an ideal is an exact functor on finite modules over Noetherian rings. This produces the right vertical isomorphism in the diagram of graded rings

$$\begin{array}{ccc}
\mathrm{Gr}_{\mathfrak{m}_s}(\mathcal{O}^\wedge_{W,w}) & \longleftarrow & \mathrm{Gr}_{\mathfrak{m}_s}(\mathcal{O}^\wedge_{(Y \times_T \tau V,(y,v)}) \\
\uparrow & & \uparrow \cong \\
\mathrm{Gr}_{\mathfrak{m}_s}(\mathcal{O}^\wedge_{X,x}) & \xrightarrow{\varphi} & \mathrm{Gr}_{\mathfrak{m}_s}(\mathcal{O}^\wedge_{Y_\sigma,y_\sigma})
\end{array}$$

We do not claim the diagram commutes. By the result of the previous paragraph the left arrow is surjective. The other three arrows are isomorphisms. It follows that the left arrow is a surjective map between isomorphic Noetherian rings. Hence it is an isomorphism by Algebra, Lemma algebra-lemma-surjective-endo-noetherian-ring-is-iso (you can argue this directly using Hilbert functions as well). In particular $\mathcal{O}^\wedge_{X,x} \to \mathcal{O}^\wedge_{W,w}$ must be injective as well as surjective which finishes the proof. □